

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **FORMAMIDA**  
 Product Description: **Formamide**  
 Cat No. : 205820000; 205820010; 205820025; 205821000  
 Synonyms Carbamaldehyde; Methanamide.  
 CAS No 75-12-7  
 Molecular Formula C H<sub>3</sub> N O

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

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**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                      |
|------------------------------------------------------|----------------------|
| Carcinogenicity                                      | Category 2 (H351)    |
| Reproductive Toxicity                                | Category 1B (H360FD) |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373)    |

**Label Elements**

**Signal Word**
**Danger**
**Hazard Statements**

H351 - Suspected of causing cancer

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H360FD - May damage fertility. May damage the unborn child  
H373 - May cause damage to organs through prolonged or repeated exposure

## Precautionary Statements

### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P308 + P313 - IF exposed or concerned: Get medical advice/attention

### Storage

P403 - Store in a well-ventilated place

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component | CAS No  | Weight % |
|-----------|---------|----------|
| Formamide | 75-12-7 | >95      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

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## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

### **Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen cyanide (hydrocyanic acid), Ammonia.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

### Environmental precautions

Should not be released into the environment.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

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## Control Parameters

| Component | Malaysia | ACGIH TLV          | OSHA PEL                                                                                                                       |
|-----------|----------|--------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Formamide |          | TWA: 1 ppm<br>Skin | (Vacated) TWA: 20 ppm<br>(Vacated) TWA: 30 mg/m <sup>3</sup><br>(Vacated) STEL: 30 ppm<br>(Vacated) STEL: 45 mg/m <sup>3</sup> |

| Component | European Union | The United Kingdom                                                                                             | Germany |
|-----------|----------------|----------------------------------------------------------------------------------------------------------------|---------|
| Formamide |                | STEL: 30 ppm 15 min<br>STEL: 56 mg/m <sup>3</sup> 15 min<br>TWA: 20 ppm 8 hr<br>TWA: 37 mg/m <sup>3</sup> 8 hr | Haut    |

## Exposure Controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

|                          |                       |
|--------------------------|-----------------------|
| Eye Protection           | Goggles               |
| Hand Protection          | Protective gloves     |
| Skin and body protection | Long sleeved clothing |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                          |                                                                                                                                                                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Respiratory Protection   | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators                                                                                                                                          |
| Recommended Filter type: | Organic gases and vapours filter Type A Brown conforming to EN14387<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly<br>When RPE is used a face piece Fit Test should be conducted |

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice

**Environmental exposure controls** No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                |                   |                |
|----------------|-------------------|----------------|
| Appearance     | Clear             |                |
| Physical State | Liquid            |                |
| Odor           | Ammonia-like      |                |
| Odor Threshold | No data available |                |
| pH             | 4-5               | 200 g/l aq.sol |

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|                     |                           |                                   |
|---------------------|---------------------------|-----------------------------------|
| Melting Point/Range | 2 - 3 °C / 35.6 - 37.4 °F |                                   |
| Softening Point     | No data available         |                                   |
| Boiling Point/Range | 210 °C / 410 °F           |                                   |
| Flash Point         | 175 °C / 347 °F           | Method - No information available |

|                          |                       |        |
|--------------------------|-----------------------|--------|
| Evaporation Rate         | No data available     |        |
| Flammability (solid,gas) | Not applicable        | Liquid |
| Explosion Limits         | Lower 2.7<br>Upper 19 |        |

|                              |                          |             |
|------------------------------|--------------------------|-------------|
| Vapor Pressure               | 0.08 mbar @ 20 °C        |             |
| Vapor Density                | 1.56                     | (Air = 1.0) |
| Specific Gravity / Density   | 1.133                    |             |
| Bulk Density                 | Not applicable           | Liquid      |
| Water Solubility             | Miscible                 |             |
| Solubility in other solvents | No information available |             |

|                                         |         |
|-----------------------------------------|---------|
| Partition Coefficient (n-octanol/water) |         |
| Component                               | log Pow |
| Formamide                               | -0.82   |

|                           |                          |
|---------------------------|--------------------------|
| Autoignition Temperature  | 500 °C / 932 °F          |
| Decomposition Temperature | 180 °C                   |
| Viscosity                 | 3.75 mPa.s at 20 °C      |
| Explosive Properties      | No information available |
| Oxidizing Properties      | No information available |

|                   |          |
|-------------------|----------|
| Molecular Formula | C H3 N O |
| Molecular Weight  | 45.04    |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### Conditions to Avoid

Excess heat. Incompatible products.

### Incompatible Materials

Acids. Bases. Strong oxidizing agents.

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## Hazardous Decomposition Products

Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen cyanide (hydrocyanic acid). Ammonia.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

| Component | LD50 Oral                 | LD50 Dermal        | LC50 Inhalation       |
|-----------|---------------------------|--------------------|-----------------------|
| Formamide | LD50 = 5577 mg/kg ( Rat ) | 17 g/kg ( Rabbit ) | >3900 ppm ( Rat ) 6 h |

##### (b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

##### (c) serious eye damage/irritation;

Based on available data, the classification criteria are not met

##### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

##### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

Not mutagenic in AMES Test

##### (f) carcinogenicity;

Category 2

Possible cancer hazard. May cause cancer based on animal data

##### (g) reproductive toxicity;

Reproductive Effects

Category 1B

Developmental Effects

May cause harm to the unborn child. Possible risk of impaired fertility.

May cause harm to the unborn child. Developmental effects have occurred in experimental animals.

Teratogenicity

Teratogenic effects have occurred in experimental animals.

##### (h) STOT-single exposure;

Based on available data, the classification criteria are not met

##### (i) STOT-repeated exposure;

Category 2

Target Organs

Liver, Kidney, Blood.

##### (j) aspiration hazard;

Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed No information available.

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**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

| Component | Freshwater Fish                                   | Water Flea                            | Freshwater Algae                                                                                   | Microtox               |
|-----------|---------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------|------------------------|
| Formamide | LC50: = 9135 mg/L, 96h static (Brachydanio rerio) | EC50: > 500 mg/L, 48h (Daphnia magna) | EC50: > 500 mg/L, 96h (Desmodesmus subspicatus)<br>EC50: > 500 mg/L, 72h (Desmodesmus subspicatus) | EC50 > 10000 mg/L 17 h |

**Persistence and degradability** Readily biodegradable  
**Persistence** Persistence is unlikely.

**Bioaccumulative potential** Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Formamide | -0.82   | No data available             |

**Mobility in soil** The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products** Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO** Not regulated

**Road and Rail Transport** Not regulated

**IATA** Not regulated

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**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**International Inventories** X = listed

| Component | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------|-----------|------|-----|-------|------|------|-------|------|----------|
| Formamide | 200-842-0 | X    | X   | X     | X    | X    | X     | X    | KE-17231 |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date** 22-Mar-2025  
**Revision Summary** Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

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## Disclaimer

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**End of Safety Data Sheet**